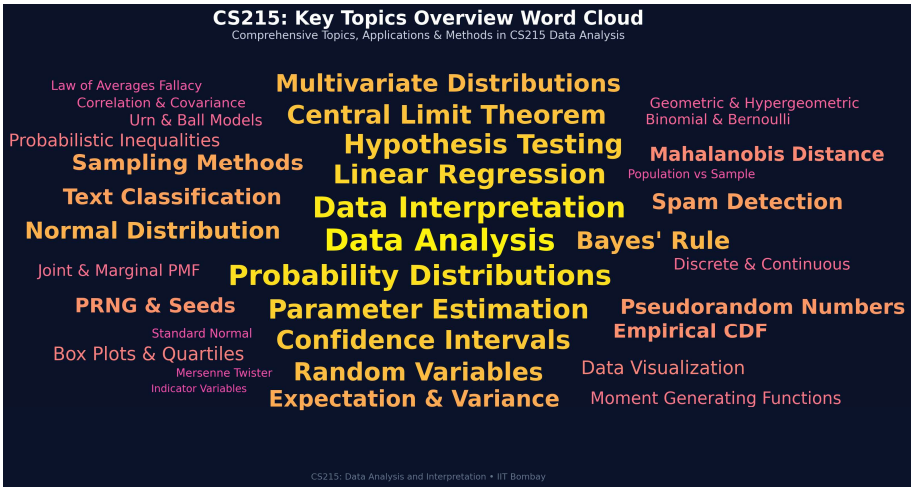
# What topics have we discussed?

- Up to chapter 6 in the textbook (Sheldon Ross)
- What is the syllabus for the exam?
  - Coming up

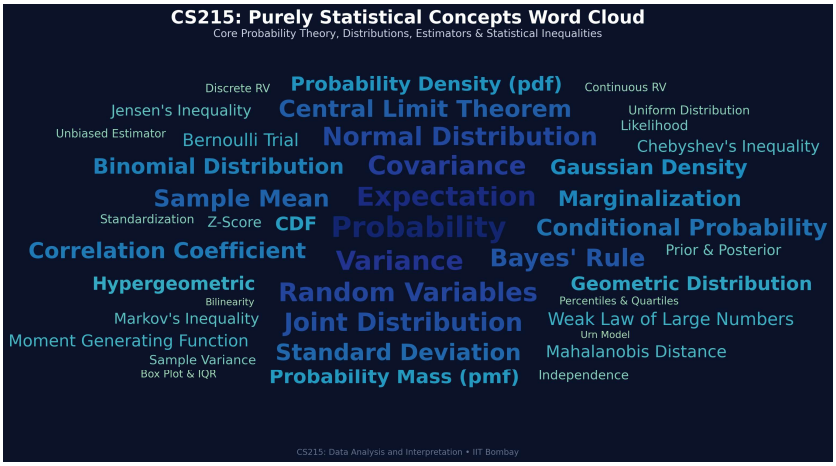
2

# We have come a long way




3

# We have come a long way




4

# Let’s revisit the first week of classes

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FOUNDATIONS

## Population

**POPULATION**

The full process or distribution

Parameters: μ and $\sigma^2$
Fixed—but usually unknown

- Suppose someone (an "oracle") told you that values are
  - Only (W= 0 or W= 2 or W= 4)
  - What's the mean and variance?
  - Expected value: $\mu = 2$
  - Variance as $\sigma^2 = 8/3$

Basic "first order" knowledge you must have

1

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FOUNDATIONS

## Population and Observations

**POPULATION**

The full process or distribution

Parameters: μ and $\sigma^2$
Fixed—but usually unknown

observe

**SAMPLE**

Observed values: $x_1, \ldots, x_n$

Statistics: $\bar{x}$ and $S^2$
Change from sample to sample

Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

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Basic "first order" knowledge you must have

# Example

- The weights of men has a mean of 70 kg with a variance of 2 kg
- Want mean and variance of the *total of 3 randomly* sampled men?
  - Assume independence, weights are normally distributed, let X be the weight of a randomly sampled man and T the total weight
  - A student wrote: T = 3X
  - Using the properties discussed in class, $E[T] =?$ $\text{Var}[T] = ?$

$$X \sim N(\mu, \sigma^2) \text{ and } Y = aX, \implies Y \sim N(a\mu, a^2\sigma^2)$$

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FOUNDATIONS

## Population and Observations

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**SAMPLE**

Observed values: $x_1, \ldots, x_n$

Statistics: $\bar{x}$ and $S^2$
Change from sample to sample

$$\overline{x} = \frac{\sum x_i}{n} \qquad \frac{\sum (x_i - \overline{x})^2}{n-1}$$

Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

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# Experiments

Focusing on the first column, how do we use the language of mathematics to express the first column

| All possible samples of size 2 | $\bar{b} = \frac{\sum x_i}{n}$ | $\frac{\sum(x-\bar{b})^2}{n-1}$ | $\frac{\sum(x-\bar{b})^2}{n}$ |
|---|---|---|---|
| 0 0 | (0+0)/2 = 0 | $((0-0)^2+(0-0)^2)/1$ = 0 | 0 |
| 0 2 | 1 | $((0-1)^2+(2-1)^2)/1$ = 2 | 1 |
| 0 4 | 2 | 8 | 4 |
| 2 0 | 1 | 2 | 1 |
| 2 2 | 2 | 0 | 0 |
| 2 4 | 3 | 2 | 1 |
| 4 0 | 2 | 8 | 4 |
| 4 2 | 3 | 2 | 1 |
| 4 4 | 4 | 0 | 0 |

- Does the first column represent
  - A: Two samples from one random variable W?
  - B: One sample each from two instances of the random variable W?

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# Experiments

Focusing on the first column, how do we use the language of mathematics to express the first column




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| 0 4 | 2 | 8 | 4 |
| 2 0 | 1 | 2 | 1 |
| 2 2 | 2 | 0 | 0 |
| 2 4 | 3 | 2 | 1 |
| 4 0 | 2 | 8 | 4 |
| 4 2 | 3 | 2 | 1 |
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FOUNDATIONS

## Population and Observations

**POPULATION**

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Fixed—but usually unknown

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Observed values: $x_1, \ldots, x_n$

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Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

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## Experiments

Focusing on the first column

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| 2 0 | 1 | 2 | 1 |
| 2 2 | 2 | 0 | 0 |
| 2 4 | 3 | 2 | 1 |
| 4 0 | 2 | 8 | 4 |
| 4 2 | 3 | 2 | 1 |
| 4 4 | 4 | 0 | 0 |

- Assume the sample was of size 1
- What is the statistician's view of the expected value of the random variable W?

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# The weak law of large numbers

- Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables, each with mean $\mu$. For any $\varepsilon > 0$:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \text{ as } n \to \infty$$

- The **sample mean** converges in probability to the true mean

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# Experiment

Now focusing on the second column

| All possible samples of size 2 | $\bar{b} = \frac{\sum x_i}{n}$ | $\frac{\sum(x-\bar{b})^2}{n-1}$ | $\frac{\sum(x-\bar{b})^2}{n}$ |
|---|---|---|---|
| 0 0 | (0+0)/2 = 0 | $((0-0)^2+(0-0)^2)/1 = 0$ | 0 |
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| 0 4 | 2 | 8 | 4 |
| 2 0 | 1 | 2 | 1 |
| 2 2 | 2 | 0 | 0 |
| 2 4 | 3 | 2 | 1 |
| 4 0 | 2 | 8 | 4 |
| 4 2 | 3 | 2 | 1 |
| 4 4 | 4 | 0 | 0 |

- Considering samples of size 2:
  - What will be statistician view of the expected value of the random variable?

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# Experiment

Now focusing on the last two columns

| All possible samples of size 2 | $\bar{b} = \frac{\sum x_i}{n}$ | $\frac{\sum(x-\bar{b})^2}{n-1}$ | $\frac{\sum(x-\bar{b})^2}{n}$ |
|---|---|---|---|
| 0 0 | (0+0)/2 = 0 | $((0-0)^2+(0-0)^2)/1 = 0$ | 0 |
| 0 2 | 1 | $((0-1)^2+(2-1)^2)/1 = 2$ | 1 |
| 0 4 | 2 | 8 | 4 |
| 2 0 | 1 | 2 | 1 |
| 2 2 | 2 | 0 | 0 |
| 2 4 | 3 | 2 | 1 |
| 4 0 | 2 | 8 | 4 |
| 4 2 | 3 | 2 | 1 |
| 4 4 | 4 | 0 | 0 |

- Considering samples of size two,
  - Dividing by n results in $E[\overline{s}^2] = \frac{12}{9} = \frac{4}{3}$
  - Division by n underestimates the true variance
  - Dividing by (n-1) results in $E[\overline{s}^2] = \frac{24}{9} = \frac{8}{3}$

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# CLT

- As stated, the random variables $X_i$ are independent and identically distributed.
- Alternate: require independence but allow the random variables to belong to different distributions
  - Technical conditions (e.g. variance of no one random variable, or groups of random variables is allowed to dominate the sum of the variances)
  - "take home message" … it works

○ Misconceptions

- "Lots" of random sampling "reveals" normal distribution
- Does not apply to random sampling of any variable
  - Applies to sampling distribution of mean
- Empirical rule of thumb (n=30) can lead to seriously flawed inferences

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